IT4010 Research Project – 2025 July Topic Assessment Form

Project ID: 25-26J-129

1. Topic (12 words max)

Modular Multi-Agent Document AI: Contextual Memory, HITL, Hallucination Detection, Guardrails

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

As AI continues to evolve, there's growing interest in building systems that can act more like intelligent assistants taking initiative, adapting to new situations, and working together to complete complex tasks. One area where this potential is especially exciting is document extraction: automatically pulling structured information from things like contracts, invoices, or medical records. But current systems still struggle with being too rigid, making mistakes, or failing to understand the full context of the documents they're processing.
Most traditional approaches treat each document as a one-off task, with no memory of past interactions or understanding of domain-specific rules. This often leads to errors like missing key details or generating incorrect information especially when documents are unclear or vary in format. These issues make it hard to trust AI for document processing in fields where precision really matters, like law, finance, or healthcare.
To address this, our research explores a smarter, more flexible approach using what's called **agentic AI,** a system made up of multiple smaller AI agents, each handling different parts of the task. For example, one agent might focus on understanding document structure, another on checking for errors, and another on making sure everything follows the correct format. These agents can work together, just like a team, to get better results.
We also bring in a kind of "memory" for the system, so it can remember previous documents and decisions. This helps reduce repeated mistakes and makes the AI more consistent over time. To ensure the system stays reliable, we include checks that spot where the AI might be making things up (called hallucinations) and allow humans to step in when needed. These human corrections can then be fed back into the system to help it improve.

In short, our goal is to build a next-generation document extraction system that's smarter, more reliable, and better suited for real-world use one that combines teamwork (through multiple AI agents), memory, human feedback, and strong safety checks.

References -

[1]Aman Raghuvanshi, "Agentic AI #3 — Top AI Agent Frameworks in 2025: LangChain, AutoGen, CrewAI & Beyond," *Medium*, May 26, 2025. https://medium.com/%40iamanraghuvanshi/agentic-ai-3-top-ai-agent-frameworks-in-2025-langchain-autogen-crewai-beyond-2fc3388e7dec

[2]"Adaptive Semantic Prompt Caching with VectorQ," *Arxiv.org*, 2023. https://arxiv.org/html/2502.03771v1?utm_source=chatgpt.com

[3]X. Wu, T. Ma, X. Li, Q. Chen, and L. He, "Human-In-The-Loop Document Layout Analysis," *arXiv.org*, 2021. https://arxiv.org/abs/2108.02095?utm_source=chatgpt.com

[4]D. Gosmar and D. A. Dahl, "Hallucination Mitigation using Agentic AI Natural Language-Based Frameworks," *arXiv (Cornell University)*, Jan. 2025, doi: https://doi.org/10.48550/arxiv.2501.13946.

[5]D. Gosmar and D. A. Dahl, "Hallucination Mitigation using Agentic AI Natural Language-Based Frameworks," *arXiv.org*, 2025. https://arxiv.org/abs/2501.13946

[6]S. Wadhwa, "Advancing Agentic AI with Smart Caching and Adaptive Multi-Agent Frameworks," *Medium*, Feb. 11, 2025. https://medium.com/@sumitwadhwa.iitr/advancing-agentic-ai-with-smart-caching-contextual-data-capture-and-adaptive-frameworks-f15a8b915da6 (accessed Jun. 22, 2025).

71]Sachin Kediyal, "THE CRITICAL ROLE OF HUMAN-IN-THE-LOOP APPROACHES FOR AI SYSTEM SUCCESS," vol. 7, no. 2, pp. 2579–2588, Dec. 2024, doi: https://doi.org/10.34218/ijrcait_07_02_194.

[8]S. Kundu, "Collaborative Agentic AI for Global Resource Management: Optimizing Sustainability and Efficiency Across Industries," *International Journal of Multidisciplinary Research and Growth Evaluation.*, vol. 5, no. 2, pp. 1023–1027, 2024, doi: https://doi.org/10.54660/.ijmrge.2024.5.2.1023-1027.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

   The proposed solution is a modular agentic AI system capable of intelligently automating document extraction in highly accurate, flexible, and secure ways. It consists of four main components working together:

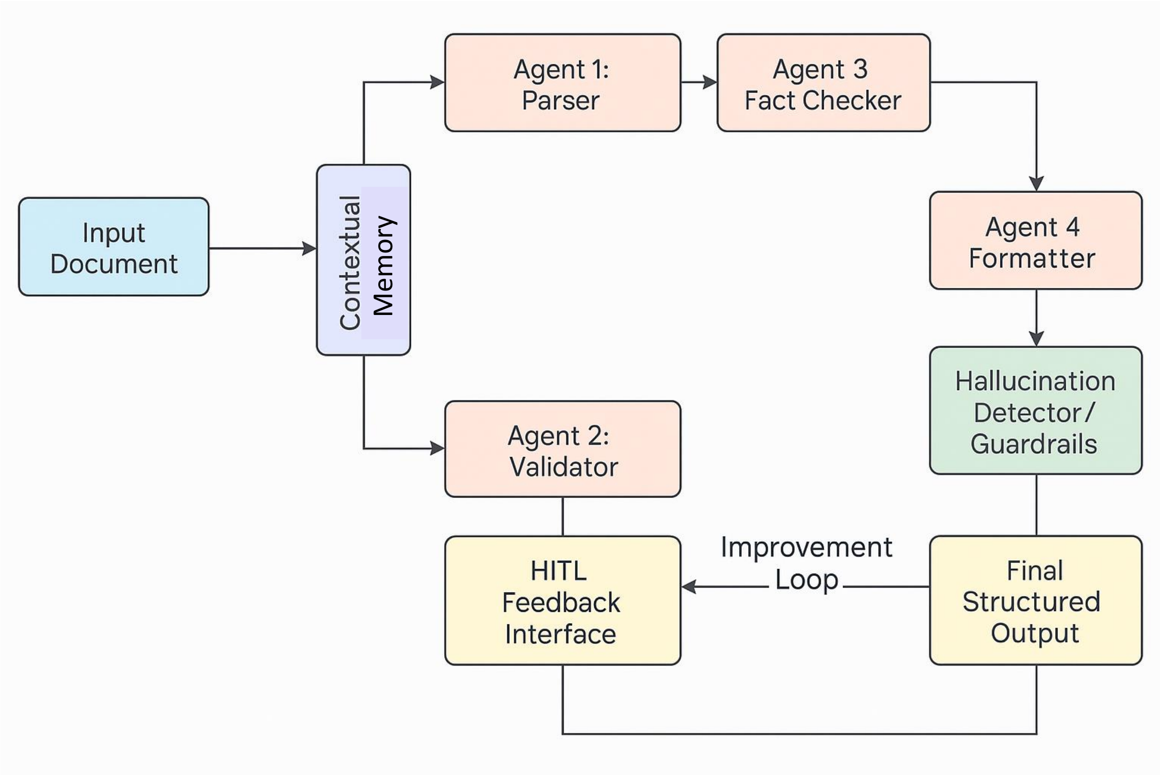
   Multi-Agent Architecture: The system consists of several specialist agents such as parser, validator, fact-checker, and summarizer that collaborate to extract information, clean and structure it from documents. The agents function in a modular, scalable pipeline and are controlled with different tools.

   Contextual Memory and Caching: A memory plane, powered by vector databases and semantic search, allows agents to recall previously experienced contextually applicable interactions or already processed material. This boosts consistency, reduces repetition, and supports adaptive long-term behavior.

   Hallucination Detection and Guardrails: For trustworthiness reasons, the system incorporates validation routines like confidence scoring and schema enforcement. These guardrails detect hallucinations or inconsistencies and either automatically fix or redirect them to human moderators.

   Human-in-the-Loop (HITL) Feedback: A user interface enables review, revising, and approval of AI output by subject matter experts. Their revisions are noted and implemented to enhance future system responses in an ongoing cycle of refinement.

   Integrating these components produces an engaged and dependable document extraction solution that can exist independently in high-stakes environments yet still allow human review if necessary.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

   Technical Competence: The architecture necessitates robust knowledge in distributed systems design, multi-agent coordination protocols, and LLM/SLM integration patterns. Expertise in vector database optimization, semantic search algorithms, and caching strategy from the contextual memory component is necessary. Workflow orchestration tool skills and real-time data pipeline design know-how are important in HITL deployment.

   AI/ML Specialization: Strong expertise in transformer architecture, prompt engineering, and model fine-tuning techniques. Great experience with hallucination detection techniques, confidence scoring models, and guardrail deployment. Knowledge of the large language model deployment strategies (GPT, Claude) and small language model deployment strategies (Phi, Gemma).

   Domain Knowledge for Document Processing: Document structure analysis, OCR integration, layout detection, and multi-modal content extraction knowledge. Understanding of various document formats (PDFs, images, structured/unstructured text) and their processing challenges. Information extraction norms and compliance standards knowledge.

Data Requirements: Huge, labeled document training and validation sets, diverse types of documents, languages, and quality. Ground truth annotations for extraction work, error classification datasets for hallucination detection model training. Historical interaction data for contextual memory optimization and user behavior pattern analysis.

Infrastructure Skills: Container orchestration (Kubernetes/Docker), cloud platform management (AWS/Azure/GCP), monitoring and observability platforms. Database administration for vector stores as well as traditional databases, API design and management, security and compliance models.

Business Domain Knowledge: Industry-specific knowledge (Financial) for document types specific to industries. Understanding regulatory requirements, data privacy laws, and industry-specific quality standards. Business process automation and change management knowledge for enterprise deployment.

Quality Assurance: Familiarity with AI system testing techniques, including adversarial testing, detection of edge cases, and continuous monitoring processes. Knowledge of human-AI collaboration behavior and feedback loop optimization.

8. Objectives and Novelty

Main Objective: The objective of this research is to develop a robust agentic AI system for document extraction that combines advanced contextual memory, proactive error detection, multi-agent collaboration, and human-in-the-loop feedback to deliver highly accurate, adaptable, and trustworthy results across diverse document types.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Wijetunga W.L.A.T. - IT22910936	**Hallucination Detection & Guardrails**: Prevent and filter out AI-generated hallucinations by combining confidence scoring, source grounding, schema constraints, and domain rules to ensure output fidelity	• Integrate multi-tiered confidence scoring (logits, entropy, ensemble) to flag low-certainty outputs • Implement retrieval-augmented grounding: fetch and cite source passages for every assertion • Define schema and regex-based guardrails for domain fields • Encode rule-based validators (e.g., numeric ranges, date formats) • Route flagged outputs to human review via HITL interface	• **Dynamic Confidence Ensemble:** Fuses multiple uncertainty signals (logit-based, calibration-powered, and ensemble votes) for robust hallucination flagging • **Proactive RAG Grounding:** Embeds on-the-fly retrieval calls into generation prompts to prevent hallucinations before they occur rather than post-hoc detection • **Schema-Driven Filtering:** Leverages JSON/YML schemas plus regex patterns to automatically validate structure and domain constraints • **Rule-Based Safe-Completion**: Applies lightweight, domain-

			specific rules (e.g., "EBITDA must be positive") to enforce hard guardrails at inference time
Dasanayake D. R. L. – IT22350800	Contextual Memory, Caching & Adaptive Responses:	• Design and implement contextual memory modules that allow agentic AI systems to capture, store, and use relevant context from both users and their environment. • Integrate both short-term memory for immediate, session-based context and long-term memory for persistent, historical knowledge, enabling agents to reason and adapt over time. • Enable semantic retrieval so agents can intelligently access and reuse information from past documents and previous agent interactions using vector embeddings and knowledge graphs, rather than relying only on keyword matching. • Develop adaptive prompt strategies that use retrieved contextual information to dynamically tailor agent responses, improving relevance, accuracy, and user alignment • Set up a feedback loop so that human corrections and expert feedback are continuously incorporated into the memory	• **Context-Aware Retrieval**: Retrieves information based on current task context and past interactions, not just keywords. • **Semantic & Associative Linking:** Connects related memories using vector embeddings, enabling recall of conceptually similar cases. • **Multi-Type Memory Integration:** Combines episodic (history), semantic (facts), and procedural (steps) memory for richer reasoning. • **Continuous Learning from Feedback:** Updates memory directly from human corrections, improving future extractions. • **Self-Organizing Structure:** Memory evolves dynamically, linking new knowledge as it's learned, without rigid schemas.

		system, allowing the AI to learn and improve with each interaction.	
Pathirana M.M - IT22335050	System Architecture & Multi-Agent Collaboration	• Define agent roles and capabilities (e.g., ExtractorAgent, ValidatorAgent, FeedbackAgent) • Implement agent communication and coordination workflows using LangChain or CrewAI • Enable task routing and dependency handling across agents • Integrate shared memory and context passing between agents • Monitor agent performance and system bottlenecks	• **Dynamic Task Routing:** Introduces intelligent routing and fallback for error handling and uncertainty management • **Multi-Agent Modularization:** Decomposes complex NLP tasks into specialized autonomous agents for parallel and efficient execution • **Scalable & Extensible Framework:** System designed to easily accommodate new agents and workflows with minimal reconfiguration
Nayanapriya S.A.T - IT22892058	Design a modular ETL pipeline & Integrate a Human-in-the-Loop (HITL)	• Implement document ingestion from various formats • Build a HITL interface for reviewing and correcting extracted fields. • Log user corrections to improve extraction heuristics over time.	• **Human-in-the-Loop Extraction:** Combines automated data extraction with human oversight to ensure accuracy and trustworthiness. • **Multi-format Document Handling:** Supports heterogeneous document types, improving flexibility and coverage. • **Modular & Scalable Design:** Designed to easily plug into

			larger LLM workflows or analytics systems.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Wijetunga W.L.A.T. - IT22910936	**Hallucination Detection & Guardrails** - Harnesses advanced data-science techniques—uncertainty quantification, retrieval-augmented generation, schema validation, and rule-based checks—to proactively prevent and detect AI hallucinations. It combines rapid ensemble scoring with inline grounding and guardrails to ensure every output is both accurate and compliant with domain constraints.
Dasanayake D. R. L. – IT22350800	**Contextual Memory for Agentic AI Systems:** Contextual memory equips agentic AI with the ability to store, retrieve, and adapt knowledge from previous tasks and interactions, directly leveraging data science skills. As a data science specialist, I design and implement both short-term (session-based) and long-term (cross-session) memory modules using vector databases and semantic retrieval. This enables agents to learn from historical data, user feedback, and correction logs, ensuring outputs are consistent, adaptive, and data driven. The approach applies core data science principles such as efficient data storage, retrieval algorithms, and feedback-based model improvement to create AI systems that are not only more intelligent and reliable, but also auditable and continuously improving. This integration is essential for building scalable, reflective, and trustworthy agentic AI solutions in real-world environments.
Pathirana M. M - IT22335050	**System Architecture & Multi-Agent Collaboration -** As a data science specialist, I design and implement modular multi-agent systems that decompose complex AI workflows into manageable, intelligent components—each responsible for a specific task such as ingestion, extraction, validation, or feedback. This architecture applies core data science principles of system modularity, task optimization, and workflow orchestration to ensure efficiency, traceability, and scalability. This architecture enables robust, interpretable AI pipelines by combining data engineering skills (workflow orchestration, modular pipeline design) with data science techniques (feedback loops, performance metrics, error detection).
Nayanapriya S.A.T - IT22892058	**ETL pipeline & Integrate a Human-in-the-Loop (HITL)** -As a data engineering specialist, I architect and implement a modular ETL pipeline augmented with Human-in-the-Loop (HITL) capabilities. The system decomposes the data ingestion and transformation process into specialized agents—each responsible for a discrete phase such as extraction, normalization, validation, or human feedback. This modular design ensures fault isolation, scalability, and ease of maintenance. The architecture leverages data engineering principles of workflow orchestration, schema validation, and modular transformation while embedding human oversight in critical stages for quality assurance. The HITL interface acts as an intelligent collaboration point between automated agents and human validators, enabling real-time correction, feedback collection, and continuous pipeline refinement. By integrating metrics

	and feedback loops, the system enhances trust, transparency, and adaptability in downstream data applications.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Samadhi	Rathnayake	
Co-Supervisor	Dr.	Lakmini	Abeywardena	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	

Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important**:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.